Programming Assignment 1

Dinosaur Game

Due Date: 2025/10/15 (Wed.) 23:59:59

- version 1 20250911
- version 2 20250919
- version 3 20250921

1. Background

Bob is a student who also signed up for Introduction to Computer I. But at the start of the semester, he thinks that the workload of this class is too heavy. He would like to have some space and time to play with everybody's favorite game, Dinosaur Game. But the regular one on the website is too boring, so he came up with the idea to write it on his own. Please help him to complete the game. In this programming assignment, you are asked to simulate a Dinosaur Game which is usually seen when no network is available on Chrome.



2. Description

In this programming assignment, you will help Bob in his quest to complete the Dinosaur Game. We transform the original continuous scene of Dinosaur Game into a discrete map so that you can use Cartesian coordinates to understand it. Fig.1 will help you to comprehend the meaning of 'discrete'.

Time in this game is also divided into discrete units called ticks. At the beginning of each tick, the dinosaur can receive a command to move.

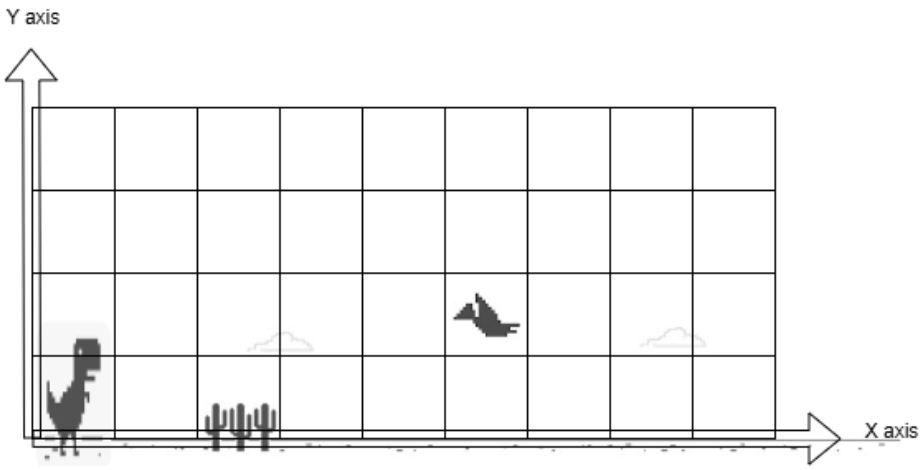


Fig.1 Discrete Dinosaur Game, we always use where the dinosaur standing as the y-axis representation. For example, the coordinate of the dinosaur standing is (0,0). The dinosaur can move based on discrete (x, y) coordinate.

There is a little difference between the regular Dinosaur Game and this project. In the regular one, the dinosaur will move forward each tick. But in this PA, we decided to make the dinosaur always stay in x=0, and make all the obstacles move 1 slot left every tick. You can image that if a car is moving right, it is equal to the entire environment is moving left.

You can control the dinosaur to perform basic actions such as *jump*, *crouch*, or *do nothing*. We encode these movement as number calls *command* so that you can enter these movements to your program conveniently. You can issue only one command per tick. These commands **require** fool-proofing. Following are the explanations of these movement. The message of fool-proofing please refer to Output section.

- Jump: There are four steps for a complete jump procedure. If the player wants to enter a jump command but the dinosaur is not yet finishing the last jumping procedure, it is considered invalid. After you enter the command of the jump at tick=0 and it is valid, the dinosaur will jump to y=1 at the first tick, y=2 at the second tick, and fall to y=1 at the third tick, and finally return to the ground (y=0) at the fourth tick.
- Crouch: If the dinosaur crouches, its height will be 1, and its position will be forced to y=0. For example, if you enter the jump command, the dinosaur's position at the first tick will be y=1. If you then enter the crouch command, the dinosaur's position at the second tick will be y=0.
- Do nothing:
  - If the dinosaur is in the procedure of jumping, it will stick to the sequence of jumping.
  - If the dinosaur was crouching, it will straighten back to height of 2 units.

Table.1 Command of the dinosaur movement

Action	Do nothing	Jump	Crouch
Command	0	1	2

Apart from the dinosaur, there are two more elements to form the game, which are *cactus* and *pterosaur*. We call these elements *obstacles*, which the dinosaur must dodge and gain a narrow escape. We guarantee that only one obstacle, either a cactus or a pterosaur, will appear in a single column.

You should notice that if the dinosaur doesn't collide with an obstacle in this tick and the next tick after command input, the dinosaur will stay alive. In other words, to check the obstacle while the dinosaur is moving is meaningless.

There are some more details to describe the dinosaur and obstacles:

- Dinosaur's width=1, height=2 if it is not crouching; Height=1 if it is.
- Cactus's width=1, height can be 1 or 2, must grow on the ground.
- Pterosaur's width=1, height=1, can appear in y=1 or y=2.

Since the player are going to enter these obstacles information into the program and the program is supposed to generate the corresponding pattern, we introduce *obstacle code* to enter these information efficiently. Notice that the obstacle is not related to the height of the obstacles. As for the output, we use *symbol* to represent the dinosaur position and the obstacles distribution. Table.2 shows the obstacle code and the symbol of the dinosaur and the obstacles.

Table.2 Obstacle code and symbol in the game

Description	Obstacle code	Symbol
The dinosaur	-	D
No obstacle	0	' '(two spaces)
Cactus (height=1)	1	' C'(1 space + 'C')
Cactus (height=2)	2	' C'(1 space + 'C')
Pterosaur (flies in y=1)	3	' P'(1 space + 'P')
Pterosaur (flies in y=2)	4	' P'(1 space + 'P')

At the beginning of the game, you will input an 8-digit number, which represents the obstacle distribution starting from the column next to the dinosaur. It is guaranteed that at tick 0, no obstacle will appear in column x=0, where the dinosaur is located.

- For example, the initial obstacle code to represent Fig.1 will be 01003000.
- After entering -1, don't print the map cause the dinosaur wins the game. Do not check whether the dinosaur still alive or not at the next tick.
- You must come up with a solution to deal with the leading 0 in the obstacle code.

Your program must both print the map showing the positions of the dinosaur and obstacles, and accept commands to control the little dinosaur. Please refer to Input and Output section of this document to get the picture of when to give the command and when to print the position map. It is time for you to make your Dinosaur Game.

3. Limitation

- Libraries exclude <iostream> is prohibited to include.
- You cannot print the sample output directly as the answers of the testcases.

4. Input

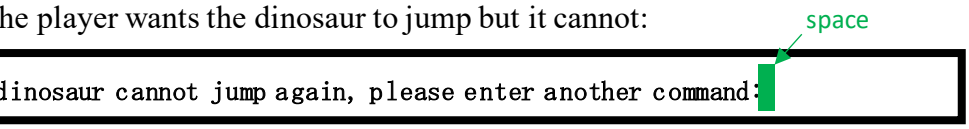
(1) Ask the player to enter 8-digit number representing the initial obstacle distribution. Input range: 0~44444444. **No** fool-proofing is needed.

```
Please enter the initial obstacle setup of the Dinosaur Game. Enter -1 to
quit:
```

(2) Enter the command to control the dinosaur. Input range: -2147483648 ~ 2147483647. This part **needs** a fool-proofing to get a correct command.

```
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch:
```

- If the player wants the dinosaur to jump but it cannot:



- If the player enters a code that dinosaur doesn't understand, for example, -1:

```
The dinosaur doesn't understand this command, try another one:
```

(3) Ask the player to enter the next round obstacle.

```
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -
1 to quit:
```

5. Output

(1) Show the welcome message at the beginning of the game.

```
=====Welcome to the Dinosaur Game=====
```

(2) Print the 3 types of objects, including *dinosaur*, *cactus*, and *pterosaur* in correct positions.

- Dinosaur shows ' D'. Notice that its height will be 2 if it is not crouching.
- Cactus shows ' C'.
- Pterosaur shows ' P'.
- If there are no obstacles, shows two spaces.
- Each column should be divided by '|'.

The following example shows the situation that initial obstacle setup code = 01234023.

```
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |P|  |  |  |
| D|  |  |C|P|  |  |C|P|
| D|  |C|C|  |  |  |C|  |
```

(3) Repeat accepting future obstacle code, dinosaur movement command and print

the game window until the game ends.

(4) At the end of the game, there are three types of messages to print depending on the situation, followed by a farewell message before the program ends.

- If player enters -1 at step of enter initial code:

```
The game ended when it just started.
=====Thanks for playing Dinosaur Game, see you next time=====
```

- If the dinosaur stays alive after -1 is entered:

```
Congratulations, you WIN the game!!!
=====Thanks for playing Dinosaur Game, see you next time=====
```

- If the dinosaur died due to run into an obstacle:

```
What a pity, it was close :(
=====Thanks for playing Dinosaur Game, see you next time=====
```

6. Samples

(1) Sample 1

Sample Input 1:

```
01003000
0
1
1
2
1
3
2
3
2
4
2
0
1
0
0
0
2
0
1
0
0
0
2
0
1
```

```
0
```

Sample Output1 Screen (Shown with Input 1):

```
=====Welcome to the Dinosaur Game=====
Please enter the initial obstacle setup of the Dinosaur Game. Enter -1 to
quit: 01003000
|  | |  |  |  |  | |  |  |
|  | |  |  |  |  | |  |  |
| D|  |  |  | P|  |  |  |
| D|  | C|  |  |  |  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 0
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 1
|  | |  |  |  |  | |  |  |
|  | |  |  |  |  | |  |  |
| D|  |  |  |P|  |  |  |  |
| D| C|  |  |  |  |  |  | C|
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 2
|  | |  |  |  |  | |  |  |
| D|  |  |  |  |  | |  |  |
| D|  |  | P|  |  |  |  | C|
| C|  |  |  |  |  |  | C| C|
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
The dinosaur cannot jump again, please enter another command: 3
The dinosaur doesn't understand this command, try another one: 2
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 3
|  | |  |  |  |  | |  |  |
|  | |  |  |  |  | |  |  |
|  |  | P|  |  |  |  | C| P|
| D|  |  |  |  |  | C| C|  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 2
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 4
|  | |  |  |  |  | |  |  |
|  | |  |  |  |  | |  |  | P|
|  | P|  |  |  |  | C| P|  |
| D|  |  |  |  |  | C| C|  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 2
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 0
|  | |  |  |  |  | |  |  |
|  | |  |  |  |  |  | P|  |
| P|  |  |  |  | C| P|  |  |
| D|  |  |  | C| C|  |  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
```

```
2=crouch: 1
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 0
|  | |  |  |  |  |  |  |
| D|  |  |  |  | P|  |  |
| D|  |  |  | C| P|  |  |
|  |  |  | C| C|  |  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 0
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 0
| D|  |  |  |  |  |  |  |
| D|  |  |  |  | P|  |  |
|  |  |  | C| P|  |  |  |
|  |  | C| C|  |  |  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 2
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 0
|  | |  |  |  |  |  |  |
|  |  |  |  | P|  |  |  |
|  |  | C| P|  |  |  |  |
| D| C| C|  |  |  |  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 0
|  | |  |  |  |  |  |  |
| D|  |  | P|  |  |  |  |
| D| C| P|  |  |  |  |  |
| C| C|  |  |  |  |  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 0
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 0
| D|  |  |  |  |  |  |  |
| D|  | P|  |  |  |  |  |
| C| P|  |  |  |  |  |  |
| C|  |  |  |  |  |  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 2
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 0
|  | |  |  |  |  |  |  |
|  | P|  |  |  |  |  |  |
| P|  |  |  |  |  |  |  |
| D|  |  |  |  |  |  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 0
What a pity, it was close :(
=====Thanks for playing Dinosaur Game, see you next time=====
```

(2) Sample 2

Sample Input 2:

```
01234023
0
1
1
4
1
1
-100
0
0
2
0
0
0
1
0
0
-1
```

Sample Output2 Screen (Shown with Input 2):

```
======Welcome to the Dinosaur Game=====
Please enter the initial obstacle setup of the Dinosaur Game. Enter -1 to
quit: 01234023
|  | |  |  |  |  |  |  |  |
|  | |  |  |  |P|  |  |  |
| D|  |  |C|P|  |  |C|P|
| D|  |C|C|  |  |  |C|  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 0
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 1
|  | |  |  |  |  |  |  |  |
|  | |  |  |  |P|  |  |  |
| D|  |C|P|  |  |C|P|  |
| D|C|C|  |  |  |C|  |C|
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 4
|  | |  |  |  |  |  |  |  |
| D|  |  |P|  |  |  |  |P|
| D|C|P|  |  |C|P|  |  |
| C|C|  |  |  |C|  |C|  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
The dinosaur cannot jump again, please enter another command: 1
The dinosaur cannot jump again, please enter another command: -100
The dinosaur doesn't understand this command, try another one: 0
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 0
```

```
| D|  |  |  |  |  |  |  |
| D|  | P|  |  |  | P|  |
| C| P|  |  | C| P|  |  |  |
| C|  |  |  | C|  | C|  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 2
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 0
|  |  |  |  |  |  |  |  |  |
|  | P|  |  |  |  | P|  |  |
| P|  |  | C| P|  |  |  |  |
| D|  |  | C|  | C|  |  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 0
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 0
|  |  |  |  |  |  |  |  |  |
| P|  |  |  |  | P|  |  |  |
| D|  | C| P|  |  |  |  |  |
| D|  | C|  | C|  |  |  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: 0
|  |  |  |  |  |  |  |  |  |
| D|  |  |  | P|  |  |  |  |
| D| C| P|  |  |  |  |  |  |
|  | C|  | C|  |  |  |  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 0
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1
to quit: -1
Congratulations, you WIN the game!!!
=====Thanks for playing Dinosaur Game, see you next time=====
```

7. Bonus

7.1 Bonus Part 1—Double Layers Dinosaur Game (10%)

After finishing the original Dinosaur Game, Bob wanted to put it online for sale. However, since commercial use might lead to copyright issues, he decided to modify the game and add more elements.

In both bonus parts, public testcase will not be one or two of the hidden testcases.

First, he plans to extend the original game, which was limited to y=3, up to y=7. Between y=3 and y=4, there is a ceiling, which may not always be complete. When the dinosaur is at the second tick of a jump (y=2) and x=1 happens to be a gap in the ceiling, the player can press jump again, it can continue jumping upward and the dinosaur can pass through the gap and move upward. The dinosaur cannot jump again until it stands firmly on the ceiling regardless of it is double jump or first jump. That is, you cannot jump again if there is a gap beneath the dinosaur.

Here is an example of jumping up to the second floor. You can discover that we use '=' to represent the ceiling. The ceiling is not counted as a Y-axis layer. The following example only demonstrate how the dinosaur jump and is not a formal output.

```
Tick 0
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
========  ==========
|  |  |  |  |  |  |  |  |  |
|  |  |  |  | P|  |  |  |  |
| D|  | P|  |  |  |  |  |  |
| D| C|  | C|  |  |  |  |  |

Tick 1
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
======  ============
|  |  |  |  |  |  |  |  |  |
| D|  |  | P|  |  |  |  |  |
| D| P|  |  |  |  |  |  |  |
| C|  | C|  |  |  |  |  |  |

Tick 2
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
====  ==============
| D|  |  |  |  |  |  |  |  |
| D|  | P|  |  |  |  |  |  |
| P|  |  |  |  |  |  |  |  |
|  | C|  |  |  |  |  |  |  |

Tick 3
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
| D|  |  |  |  |  |  |  |  |
=  ================
| D|  |  |  |  |  |  |  |  |
|  | P|  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
| C|  |  |  |  |  |  |  |  |

Tick 4
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
| D|  |  |  |  |  |  |  |  |
| D|  |  |  |  |  |  |  | C|
```

```
====================
|  |  |  |  |  |  |  |  |  |
| P|  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |

Tick 5
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
| D|  |  |  |  |  |  |  |  |
| D|  |  |  |  |  |  | C|  |
====================
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
```

The dinosaur can stand on the ceiling, but if it steps onto a gap in the ceiling, it will fall down. For example, if the dinosaur is standing at (0,4) on a solid ceiling but there is a gap at x=1, then on the next tick it will move to (0,3), then to (0,2) in the following tick, and so on until it falls back to y=0 (Don't forget the map is continuously moving left and dinosaur is also constantly moving right, so the position will always be 0). During the falling procedure, the player cannot press jump. If there is no gap for the dinosaur to cross, the second jump is not accepted, also print the same error message as in the original Dinosaur Game (The one "The dinosaur cannot jump again…").

Additionally, if the dinosaur is at y=4 or above, and the ceiling at x=1 has no gap, pressing crouch will force its position back to y=4, meaning it will return to the ceiling height. If there is a gap in the ceiling at x=1, then crouching will make it fall back to y=0 in the next tick, which is different from falling tick by tick.

As always, when determining whether the dinosaur dies, only its final position needs to be checked for obstacles. Any obstacles it might have passed through during falling can be ignored.

As part of the updated input, two additional elements will be provided. The first is the obstacle distribution on the second floor, which follows the same obstacle code rules as the first floor except for the difference in the y-axis, the obstacle codes remain unchanged but no longer need to accept -1 as end code. Here I provide a bonus opportunity for students who have carefully read this document. Please fill up the form with the fourth link provided at reference of this document by September seventeen. The passcode is hidden at somewhere in this document. Then you will get a bonus. Please do not directly tell others about this Easter egg so they can enjoy the hunting.

If the -1 is entered as the initial obstacle code for first layer, the program should print the end game and farewell message, then terminate immediately without accepting the second-floor obstacle code. The second is the completeness of the ceiling, where 0

indicates a gap and 1 indicates no gap. These three pieces of information will be entered together—initially as an 8-digit input (9-digit for ceiling distribution), and then as a 1-digit input for each subsequent tick. Another point you should pay attention is that no fool-proofing is required when entering obstacle code, but necessary when entering dinosaur movement command.

Here is an example for Bonus Part 1—Double Layer Dinosaur Game:

```
=====Welcome to the Dinosaur Game=====
Please enter the first layer initial obstacle setup of the Dinosaur Game.
Enter -1 to quit: 01234023
Please enter the second layer initial obstacle setup of the Dinosaur Game:
00000000
Please enter the initial distribution of gap in the ceiling, 0 for gap, 1
for complete ceiling: 000000000
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
=  =  =  =  =  =  =  =  =  =
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |P |  |  |  |
| D|  |  | C| P|  |  | C| P|
| D|  | C| C|  |  |  | C|  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 0
Please enter the next round's first layer obstacle setup of the Dinosaur
Game. Enter -1 to quit: 1
Please enter the next round's second layer obstacle setup of the Dinosaur
Game: 0
Please enter the next round's ceiling or gap status: 1
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
=  =  =  =  =  =  =  =  ====
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |P |  |  |  |  |
| D|  | C| P|  |  | C| P|  |
| D| C| C|  |  |  | C|  | C|
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
Please enter the next round's first layer obstacle setup of the Dinosaur
Game. Enter -1 to quit: 2
Please enter the next round's second layer obstacle setup of the Dinosaur
Game: 4
Please enter the next round's ceiling or gap status: 1
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  | P|
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
=  =  =  =  =  =  =  ======
```

```
|  |  |  |  |  |  |  |  |  |
| D|  |  | P|  |  |  |  |  |
| D| C| P|  |  | C| P|  | C|
| C| C|  |  |  | C|  | C| C|
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
The dinosaur cannot jump again, please enter another command: 0
Please enter the next round's first layer obstacle setup of the Dinosaur
Game. Enter -1 to quit: 4
Please enter the next round's second layer obstacle setup of the Dinosaur
Game: 4
Please enter the next round's ceiling or gap status: 1
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  | P| P|
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
=  =  =  =  =  =  ========
| D|  |  |  |  |  |  |  |  |
| D|  | P|  |  |  |  |  | P|
| C| P|  |  | C| P|  | C|  |
| C|  |  |  | C|  | C| C|  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
Please enter the next round's first layer obstacle setup of the Dinosaur
Game. Enter -1 to quit: 1
Please enter the next round's second layer obstacle setup of the Dinosaur
Game: 1
Please enter the next round's ceiling or gap status: 0
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  | P| P|  |
|  |  |  |  |  |  |  |  |  |
| D|  |  |  |  |  |  |  | C|
=  =  =  =  =  ========  =
| D|  |  |  |  |  |  |  |  |
|  | P|  |  |  |  |  | P|  |
| P|  |  | C| P|  | C|  |  |
|  |  |  | C|  | C| C|  | C|
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
The dinosaur cannot jump again, please enter another command: 0
Please enter the next round's first layer obstacle setup of the Dinosaur
Game. Enter -1 to quit: 0
Please enter the next round's second layer obstacle setup of the Dinosaur
Game: 0
Please enter the next round's ceiling or gap status: 1
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  | P| P|  |  |
| D|  |  |  |  |  |  |  |  |
| D|  |  |  |  |  |  | C|  |
=  =  =  =  ========  ====
|  |  |  |  |  |  |  |  |  |
| P|  |  |  |  |  | P|  |  |
|  |  | C| P|  | C|  |  |  |
```

```
|  |  | C|  | C| C|  | C|  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 2
Please enter the next round's first layer obstacle setup of the Dinosaur
Game. Enter -1 to quit: 0
Please enter the next round's second layer obstacle setup of the Dinosaur
Game: 0
Please enter the next round's ceiling or gap status: 0
|  |  |  |  |  |  |  |  |  |
|  |  |  |  | P| P|  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  | C|  |  |
=  =  =  ========  ====  =
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  | P|  |  |  |
|  | C| P|  | C|  |  |  |  |
| D| C|  | C| C|  | C|  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 2
Please enter the next round's first layer obstacle setup of the Dinosaur
Game. Enter -1 to quit: -1
Congratulations, you WIN the game!!!
=====Thanks for playing Dinosaur Game, see you next time=====
```

7.2 Bonus Part 2—Auto Play System (10%)

After completing the Double Layer Dinosaur Game, you are allowed to proceed with this Bonus section. In this part, the rules of Double Layer Dinosaur Game are applied, including the double layer of the scene. Your program must automatically generate the commands that guide the dinosaur to avoid obstacles, instead of relying on manual input. Since the program can automatically generate the command, error command is not acceptable and no fool-proofing is needed.

The solution is not unique—as long as the dinosaur can successfully win the game. We guarantee that at least one valid solution exists. If an impossible situation arises on the first floor, sufficient time will be provided for the dinosaur to move up to the second floor and take temporary refuge. As usual, the obstacle distribution will be provided by the testcases. The passcode is wakuwaku. The example for Bonus Part 2 output will be totally the same as the Part 1 output, the only difference is the dinosaur movement command will be generated by you and the error command is removed.

The Part 2 of Bonus will be checked by a checker we will provide later to check if your movement command is legal or not and to check your output pattern meets our expectation. The checker will be the scoring standard but not Online Judge in this part. You cannot directly print the example as one of the testcase's answer because in this Bonus Part 2, hidden testcase will not include the sample case.

Here is an example totally same as the one in the Bonus Part 1 section. But notice the red word on the example must generate by your bonus program instead of entering

it manually.

The time limitation is 5 second for each testcase. At most 1,0000 number of obstacle codes and ceiling distribution code will be input. The method of calculating the program runtime will append on the tutorial of executing checker.

```
======Welcome to the Dinosaur Game======
Please enter the first layer initial obstacle setup of the Dinosaur Game.
Enter -1 to quit: 01234023
Please enter the second layer initial obstacle setup of the Dinosaur Game:
00000000
Please enter the initial distribution of gap in the ceiling, 0 for gap, 1
for complete ceiling: 000000000
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
=  =  =  =  =  =  =  =  =  =
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |P |  |  |  |
| D|  |  |C |P |  |  |C |P |
| D|  |C |C |  |  |  |C |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 0
Please enter the next round's first layer obstacle setup of the Dinosaur
Game. Enter -1 to quit:  1
Please enter the next round's second layer obstacle setup of the Dinosaur
Game: 0
Please enter the next round's ceiling or gap status: 1
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
=  =  =  =  =  =  =  =  ====
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |P |  |  |  |  |
| D|  |C |P |  |  |C |P |  |
| D|C |C |  |  |  |C |  |C |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
Please enter the next round's first layer obstacle setup of the Dinosaur
Game. Enter -1 to quit: 2
Please enter the next round's second layer obstacle setup of the Dinosaur
Game: 4
Please enter the next round's ceiling or gap status: 1
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |P |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
=  =  =  =  =  =  =  ======
|  |  |  |  |  |  |  |  |  |
| D|  |  |P |  |  |  |  |  |
| D|C |P |  |  |C |P |  |C |
```

Generate automatically

Generate automatically

```
| C| C|  |  |  | C|  | C| C|
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 0
Please enter the next round's first layer obstacle setup of the Dinosaur
Game. Enter -1 to quit: 4
Please enter the next round's second layer obstacle setup of the Dinosaur
Game: 4
Please enter the next round's ceiling or gap status: 1
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  | P| P|
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
=  =  =  =  =  =  ========
| D|  |  |  |  |  |  |  |  |
| D|  | P|  |  |  |  |  | P|
| C| P|  |  | C| P|  | C|  |
| C|  |  |  | C|  | C| C|  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
Please enter the next round's first layer obstacle setup of the Dinosaur
Game. Enter -1 to quit: 1
Please enter the next round's second layer obstacle setup of the Dinosaur
Game: 1
Please enter the next round's ceiling or gap status: 0
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  | P| P|  |
|  |  |  |  |  |  |  |  |  |
| D|  |  |  |  |  |  |  | C|
=  =  =  =  =  ========  =
| D|  |  |  |  |  |  |  |  |
|  | P|  |  |  |  | P|  |  |
| P|  |  | C| P|  | C|  |  |
|  |  |  | C|  | C| C|  | C|
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 0
Please enter the next round's first layer obstacle setup of the Dinosaur
Game. Enter -1 to quit: 0
Please enter the next round's second layer obstacle setup of the Dinosaur
Game: 0
Please enter the next round's ceiling or gap status: 1
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  | P| P|  |  |
| D|  |  |  |  |  |  |  |  |
| D|  |  |  |  |  |  | C|  |
=  =  =  =  ========  ====
|  |  |  |  |  |  |  |  |  |
| P|  |  |  |  |  | P|  |  |
|  |  | C| P|  | C|  |  |  |
|  |  | C|  | C| C|  | C|  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 2
Please enter the next round's first layer obstacle setup of the Dinosaur
Game. Enter -1 to quit: 0
```

Generate automatically

Generate automatically

Generate automatically

```
Please enter the next round's second layer obstacle setup of the Dinosaur
Game: 0
Please enter the next round's ceiling or gap status: 0
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |P|P|  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |C|  |  |
=  =  =  =========  ====  =
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |P|  |  |  |
|  |C|P|  |C|  |  |  |  |
|D|C|  |C|C|  |C|  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 2
Please enter the next round's first layer obstacle setup of the Dinosaur
Game. Enter -1 to quit: -1
Congratulations, you WIN the game!!!
=====Thanks for playing Dinosaur Game, see you next time=====
```

Generate automatically

8. Submission Requirement

Please submit a source code file (not the entire project) named StudID_PA1.cpp (e.g., 9862534_PA1.cpp) and a report named StudID_Name_PA1_report.pdf (e.g., 9862534_陳聿廣_PA1_report.pdf). If you implement a bonus version, please submit it separately from the original version. Name the source code file of the bonus 1 version StudID_PA1_bonus1.cpp (e.g., 9862534_PA1_bonus1.cpp) and file of the bonus 2 version StudID_PA1_bonus2.cpp (e.g., 9862534_PA1_bonus2.cpp) and upload it to ee-class as a separate version from the original. Note that the only acceptable report file format is .pdf, **no .doc/.docx or other files are acceptable. BE SURE to follow the naming rule mentioned above. Otherwise, your program will be not graded.**

We don't restrict the report format and length. In your report, you have to at least describe:

- How to compile and execute your program; (You can use a screenshot to explain. In this PA, it can be done in one click)
- The degree of completion of the assignment; (If you complete all requirements, just specify all)
- The hardness of this assignment and how you overcome it
- Any suggestions about this programming assignment?
- Briefly explain how you completed the Bonus Part, if you did. Your workflow or only a simple description of your logic are all accepted.

You can also put anything related to the PA in your report, such as pseudocode, control flow diagram, programming developing thought, etc.

Your program will be judged by our On-line Judge (OJ) system and checker written by TA 劉廷允. Another 4 documents will be released to explain how to access

OJ and how to use checker on workstation soon. The machine information are as follows:

- OS: Ubuntu 20.04 LTS
- OJ Version：Hydro OJ v2.33.13
- Compiler：GNU GCC 9.3.0
- Supporting C++ version：C++ 11

9. Grading

The grading is as follows:

- Correctness of your code: 50%
- Readability of your code: 10%
- The report: 10%
- Demo session: 30%
- Bonus (at most): 20%

Please submit your assignment on time. Otherwise, the penalty rule will apply:

- Within 72hrs delay: 20% off
- More than 3 days: 0 point

Be sure to attend a demo session (the time will be announced later). If you have questions, please E-mail to both me (andyygchen@ee.ncu.edu.tw) and TA 劉廷允 (tim0974040@gmail.com). We will be happy to help you.

Reference

[1] Chapter 2 ~ Chapter 4, Paul Deitel and Harvey Deitel, “C++ how to program late objects version,” 7th edition, Person 2011.

[2] https://reurl.cc/0WlkMM

[3] https://reurl.cc/DOLYrN

[4] https://reurl.cc/GNLg2d

[5] https://reurl.cc/bmaep6